

CHAPTER 5: MORPHOLOGY OF FLOWERING PLANTS

 NEET / CBSE Class 11

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1. Historical Perspective: Natural History to Physiology

The description of diverse life forms on Earth began through naked-eye observation and later magnifying tools/microscopes. Initial biology was largely **Natural History**—focusing on gross structural features (external & internal).

Reductionist Biology: Detailed morphological descriptions provided the framework for later reductionist biology, where living processes drew more attention, helping frame vital research questions in plant physiology and evolution.

Morphology:

Study of external form & structure.

Anatomy:

Study of internal structural organization.

2. Scientist Focus: Dr. Katherine Esau (1898–1997)

NCERT Highlight

[Photo:
Katherine
Esau]

Born: Ukraine (1898). Studied agriculture in Russia & Germany; Doctorate in US (1931).

Key Discovery: Reported that the *curly top virus* spreads through plants via the food-conducting phloem tissue.

Landmark Publications:

> Plant Anatomy (1954) - Revived structural plant biology worldwide. > Anatomy of Seed Plants (1960) - Revered as the "*Webster's of Plant Biology*".

Honors: 6th woman elected to National Academy of Sciences (1957); National Medal of Science (1989).

CHAPTER 5 STRUCTURAL FRAMEWORK & TABLE OF CONTENTS

5.1 The Root

5.2 The Stem

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MORPHOLOGY OF FLOWERING PLANTS — CONCEPT MAP

Hierarchical Overview

VEGETATIVE ORGANS

- **Root:** Tap, Fibrous, Adventitious
- **Stem:** Nodes, Internodes, Buds
- **Leaf:** Base, Petiole, Lamina

INFLORESCENCE & FLOWER

- **Inflorescence:** Racemose, Cymose
- **Symmetry:** Actinomorphic, Zygomorphic
- **Position:** Hypo-, Peri-, Epigynous

FLORAL WHORLS

- **Calyx & Corolla:** Aestivation types
- **Androecium:** Cohesion & Adhesion
- **Gynoecium:** Placentation types

FRUIT, SEED & TAXONOMY

- **Fruit:** Drupe (Pericarp parts)
- **Seed:** Dicot vs Monocot
- **Family:** Solanaceae description

5.1 Root Types & Origin

The root is the underground non-green axis developing from the **radicle** of the embryo.

1. Tap Root System

Direct elongation of radicle forms primary root + lateral branches (secondary, tertiary).

Example: Mustard plant (Dicotyledons).

2. Fibrous Root System

Primary root is short-lived; replaced by numerous roots originating from the base of stem.

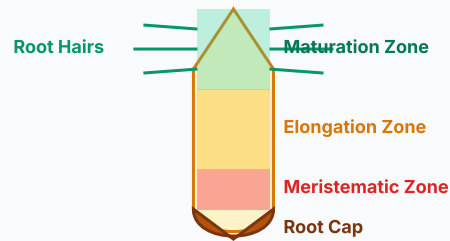
Example: Wheat plant (Monocotyledons).

3. Adventitious Root System

Roots arising from parts of the plant *other than the radicle*.

Examples: Grass, *Monstera*, Banyan tree.

5.1.1 Regions of the Root Tip



1. **Root Cap:** Thimble-like structure protecting tender root apex.
2. **Meristematic Region:** Few mm above cap; small, thin-walled, dense protoplasm cells undergoing rapid cell division.
3. **Region of Elongation:** Proximal to meristem; rapid elongation/enlargement responsible for root length growth.
4. **Region of Maturation:** Cells differentiate & mature. Epidermal cells form fine, delicate thread-like **Root Hairs** (absorb water & minerals).

Functions of the Root System

- ✓ **Absorption:** Water and mineral intake from soil.
- ✓ **Anchorage:** Provides proper support and firm attachment to plant parts.
- ✓ **Storage:** Stores reserve food materials.
- ✓ **PGR Synthesis:** Synthesizes Plant Growth Regulators (hormones).

💡 🗨 Easy Way to Remember Root Zones

"Cap Makes Every Mushroom Hairy"

Root Cap → Meristematic → Elongation → Maturation → Root Hairs (Apex to Base order).

5.2 The Stem Structure & Function

Origin: Ascending part of axis developing from the **plumule** of the embryo of a germinating seed.

Key Characteristics:

- Bears **nodes** (where leaves arise) and **internodes** (portions between nodes).
- Bears **buds**: terminal (apical) or axillary.
- Generally green when young, often becomes woody & dark brown later.

Main Functions:

1. Spreading out branches bearing leaves, flowers & fruits.
2. Conducts water, minerals, and photosynthates.
3. Food storage, support, protection, and vegetative propagation.

5.3 Structure of a Typical Leaf

Lateral, flattened structure borne at node; bears axillary bud in its axil (develops into branch). Originates from shoot apical meristem; arranged in **acropetal order**.

1. Leaf Base: Attaches leaf to stem; may bear 2 lateral small leaf-like structures called **stipules**.

Monocots: Leaf base expands into sheath covering stem.

Legumes: Swollen leaf base called **pulvinus**.

2. Petiole: Holds leaf blade to light. Long, thin, flexible petioles allow blades to flutter in wind, cooling leaf & bringing fresh air.

3. Lamina (Leaf Blade): Green expanded part with veins and veinlets. Central prominent vein = **midrib**. Veins provide rigidity & transport channels.

Venation & Types of Leaves

5.3.1 Venation Patterns

Reticulate

Veinlets form a network. Characteristic of **Dicotyledons**.

Parallel

Veins run parallel within lamina. Characteristic of **Monocotyledons**.

5.3.2 Simple vs Compound Leaves

Feature	Pinnately Compound	Palmately Compound
Leaflets Axis	Present on common axis called rachis (midrib).	Attached at a common point (tip of petiole).
Example	Neem	Silk Cotton

* Note: Axillary bud is present in axil of petiole of both simple & compound leaves, but NOT in axil of leaflets.

5.3.3 PHYLLOTAXY (Pattern of Leaf Arrangement on Stem/Branch)

High Yield Exam Topic

1. Alternate Phyllotaxy

A single leaf arises at each node in an alternate manner.

Examples: China rose, Mustard, Sunflower.

2. Opposite Phyllotaxy

A pair of leaves arises at each node and lies opposite to each other.

Examples: *Calotropis*, Guava plants.

3. Whorled Phyllotaxy

More than two leaves arise at a node and form a whorl/circle.

Example: *Alstonia*.

5.4 Inflorescence: Racemose vs Cymose

A flower is a modified shoot where shoot apical meristem transforms to floral meristem. Axis condenses, internodes do not elongate. **Inflorescence** is the arrangement of flowers on the floral axis.

Feature	Racemose Inflorescence	Cymose Inflorescence
Growth of Axis	Main axis continues to grow indefinitely.	Main axis terminates in a flower (limited growth).
Flower Order	Flowers borne laterally in Acropetal succession (youngest at apex).	Flowers borne in Basipetal order (youngest at base).

Flower Basics & Whorls

Reproductive unit in angiosperms. Arranged on swollen end of pedicel called **thalamus / receptacle**.

4 Floral Whorls:

- Accessory Organs: **Calyx** & **Corolla**
- Reproductive Organs: **Androecium** & **Gynoecium**

Perianth: Calyx & corolla not distinct (e.g., **Lily**). Members called *tepals*.

Sexuality: **Bisexual** (both stamens & carpels) vs **Unisexual** (either stamens or carpels).

Flower Symmetry & Appendages

Actinomorphic (Radial Symmetry) $\oplus$: Divided into 2 equal radial halves in ANY radial plane.

Examples: Mustard, Datura, Chilli.

Zygomorphic (Bilateral Symmetry) $\%$: Divided into similar halves ONLY in ONE vertical plane.

Examples: Pea, Gulmohur, Bean, *Cassia*.

Asymmetric (Irregular): Cannot be divided into 2 similar halves by any plane.

Example: *Canna*.

Floral Appendages: Trimerous (3s), Tetramerous (4s), Pentamerous (5s).

Bracts: Reduced leaf at base of pedicel → *Bracteate* (Br) vs *Ebracteate*.

Position on Thalamus

1. Hypogynous (Superior Ovary) $\underline{G}$

Gynoecium occupies highest position; other parts situated below it.

Examples: Mustard, China rose, Brinjal.

2. Perigynous (Half-Inferior Ovary) $\underline{G-}$

Gynoecium in centre; other parts on rim of thalamus at same level.

Examples: Plum, Rose, Peach.

3. Epigynous (Inferior Ovary) $\tilde{G}$

Margin of thalamus grows upward enclosing ovary completely; other parts arise above.

Examples: Guava, Cucumber, Ray florets of Sunflower.

SECTION 5.5.1: PARTS OF A FLOWER & PLACENTATION

NCERT Pages 63–65

5.5.1.1 Calyx & 5.5.1.2 Corolla (Aestivation)

Calyx: Sepals (Gamosepalous = united, Polysepalous = free).

Corolla: Petals (Gamopetalous = united, Polypetalous = free).

Aestivation Types (Arrangement in bud stage):

Valvate:

Margins just touch without overlapping.

Example: Calotropis.

Twisted:

One margin overlaps next appendage sequentially.

China rose, Lady's finger, Cotton.

Imbricate:

Margins overlap irregularly (no particular direction).

Cassia, Gulmohur.

Vexillary (Papilionaceous):

5 petals: Largest **Standard** overlaps 2 lateral **Wings**, which overlap 2 smallest anterior **Keels**. *Pea, Bean.*

5.5.1.3 Androecium (Stamens)

Male reproductive organ. Stamen = **Filament** + **Anther** (bilobed, 2 pollen sacs per lobe). Sterile stamen = **Staminode**.

Adhesion (Attachment to other whorls):

- **Epipetalous:** Stamens attached to petals (e.g., **Brinjal**).
- **Epiphyllous:** Stamens attached to perianth (e.g., **Lily**).

Cohesion (Fusion among stamens):

- **Monadelphous:** United into 1 bundle (e.g., **China rose**).
- **Diadelphous:** United into 2 bundles (e.g., **Pea**).
- **Polyadelphous:** United into >2 bundles (e.g., **Citrus**).
- **Polyandrous:** Stamens remain completely free.

Length Variation: Variable filament length seen in *Salvia* and *Mustard*.

5.5.1.4 Gynoecium & Placentation Types

Carpel = Stigma + Style + Ovary | Apocarpous (Free) vs Syncarpous (Fused)

Marginal

Placenta forms ridge along ventral suture; ovules in 2 rows.

Example: Pea

Axile

Placenta axial; ovules attached in multilocular ovary.

China rose, Tomato, Lemon

Parietal

Ovules on inner wall / peripheral part. False septum forms 2 chambers.

Mustard, Argemone

Free Central

Ovules borne on central axis; septa completely absent.

Dianthus, Primrose

Basal

Placenta develops at base of ovary; single ovule attached.

Sunflower, Marigold

5.6 The Fruit & Pericarp

Mature/ripened ovary post-fertilisation. **Parthenocarpic fruit** = formed without fertilisation.

Drupe Fruit (Mango & Coconut):

Develops from monocarpellary superior ovaries; 1-seeded.

- **Mango:** Thin epicarp + Fleshy edible mesocarp + Stony hard endocarp.
- **Coconut:** Mesocarp is **fibrous**.

5.7 Structure of Dicot & Monocot Seeds

Dicot Seed (e.g., Gram, Pea):

Seed coat = Outer **Testa** + Inner **Tegmen**. Scar = **Hilum**, Pore = **Micropyle**. Endospermic (Castor) vs Non-endospermic (Bean, Gram, Pea).

Monocot Seed (e.g., Maize):

Membranous seed coat fused with fruit wall. Outer protein layer = **Aleurone layer**. Shield-shaped cotyledon = **Scutellum**. Coverings: **Coleoptile** (plumule) & **Coleorrhiza** (radicle).

5.8 Floral Symbols Key

Br: Bracteate

C: Corolla

A: Androecium

G: Superior Ovary

⊕: Actinomorphic

K: Calyx

P: Perianth

G: Gynoecium

G̅: Inferior Ovary

%: Zygomorphic

Mustard Floral Formula: $\text{Br} \oplus \text{K}_{2+2} \text{C}_4 \text{A}_{2+4} \underline{\text{G}}_{(2)}$

🍷 5.9 FAMILY SOLANACEAE ('Potato Family')

⊕ ♂ K₍₅₎ C₍₅₎ A₅ G₍₂₎

Vegetative Features:

- Herbs, shrubs, small trees.
- Underground stem in potato (*Solanum tuberosum*).
- Leaves alternate, simple, exstipulate, reticulate.

Floral Features:

- Inflorescence: Solitary, axillary or cymose.
- Calyx: Sepals 5, united, persistent, valvate.
- Corolla: Petals 5, united, valvate.
- Gynoecium: Bicarpellary, syncarpous, **swollen placenta** with many ovules.

Economic Importance: Food (Tomato, Brinjal, Potato), Spice (Chilli), Medicine (*Belladonna*, *Ashwagandha*), Fumigatory (Tobacco), Ornamental (Petunia).

- **Tap root:** Direct elongation of radicle (Mustard). **Fibrous:** Base of stem (Wheat).
- **Root cap:** Protects root tip. **Root hairs:** Region of maturation.
- **Phyllotaxy:** Alternate (China rose), Opposite (Calotropis), Whorled (Alstonia).
- **Racemose:** Acropetal, unlimited growth. **Cymose:** Basipetal, main axis terminates in flower.
- **Hypogynous (Superior):** Mustard. **Perigynous (Half-inferior):** Rose, Peach. **Epigynous (Inferior):** Guava.
- **Vexillary Aestivation:** Standard (1) + Wings (2) + Keel (2). Pea/Bean.
- **Placentation:** Marginal (Pea), Axile (Lemon), Parietal (Mustard), Free Central (Dianthus), Basal (Sunflower).
- **Drupe Fruit:** Mango (edible mesocarp), Coconut (fibrous mesocarp).